The subsurface habitability of small, icy exomoons

(https://www.aanda.org/articles/aa/full_html/2020/04/aa37035-19/aa37035-19.html)

The abstract delves into the intriguing possibility that icy moons possess the necessary conditions to support life, making them valuable targets in the search for extraterrestrial existence. The premise put forth is that if exomoons are more numerous than exoplanets and have a comparable likelihood of being habitable, they could represent an extensive portion of viable environments for life throughout the Universe. The study specifically investigates the circumstances under which small, icy moons can maintain subsurface oceans, with a particular focus on the role of tidal heating as a critical factor. This process can generate enough heat to keep water in a liquid state, even when these moons are situated outside the conventional habitable zones typically defined by the presence of liquid water on the surface.

To conduct this investigation, the researchers employ a carefully structured phenomenological approach to tidal heating. They calculate the average thermal flux experienced by these moons based on their orbital dynamics and the illumination they receive from both their parent star and nearby planets, taking into account both thermal radiation and reflected light. The results from this analysis illuminate a significant relationship between the physical characteristics of a moon—such as its size, density, and orbital distance—and the depth at which ice can melt to create a subsurface ocean. This relationship is crucial for determining whether a moon can sustain liquid water beneath its icy crust.

The findings of the study are compelling: they confirm the existence of a subsurface ocean on Enceladus, one of the most intriguing moons of Saturn, known for its geysers that erupt water vapor and ice particles. Furthermore, the research indicates that Titania and Oberon, the two largest moons of Uranus, may also have conditions favorable for sustaining subsurface oceans. In contrast, the model suggests that Rhea, another of Saturn's moons, likely does not harbor liquid water, potentially limiting its habitability.

In conclusion, the paper posits that habitable environments may be present not only in Earth-like exoplanets but also in various small icy moons scattered throughout different exoplanetary systems. This extended definition of habitability could substantially broaden the scope of future astronomical observations and missions aimed at uncovering signs of life beyond our planet. The introduction emphasizes the ongoing quest to confirm the existence of such exomoons and examines how tidal heating contributes to their potential for habitability, referencing a range of previous studies that have contributed to our understanding of the conditions necessary to support life in these icy worlds.

Europa Clipper mission: the habitability of an icy moon

(https://ieeexplore.ieee.org/abstract/document/7118943)

Europa, the fourth largest moon of Jupiter, stands out as a leading candidate in the quest for extraterrestrial life due to its presumed subsurface ocean, which is believed to contain saltwater beneath a dynamic and relatively thin icy shell. This unique feature positions Europa as one of the most compelling targets in planetary science, as highlighted by the 2011 Planetary Decadal Survey, "Vision and Voyages," which emphasizes the moon's potential for supporting life.

The Europa Clipper mission is designed to conduct a comprehensive exploration of Europa, focusing on the fundamental elements essential for life: liquid water, organic and inorganic chemistry, and energy sources. By executing multiple flybys of Europa, the mission intends to study its ice shell, the interface between the ice and the liquid ocean, as well as the moon's surface geology and composition. Understanding these aspects will provide vital insight into Europa's internal processes and its habitability.

Europa's surface has been identified as sparse in craters, indicating a young and frequently renewing landscape, which suggests active geological processes that facilitate the exchange of materials between the surface and the ocean beneath. Observations from the Galileo Mission, alongside magnetic field measurements, support the theory of a subsurface ocean that could be many tens of kilometers deep, lying beneath an ice shell that ranges from a few to several tens of kilometers thick. This ocean likely comes into contact with a rocky seafloor, which may be supported by hydrothermal activity, creating a potentially rich environment for prebiotic chemistry or microbial life.

Recent data from the Hubble Space Telescope have provided preliminary evidence, albeit unconfirmed, of water and ice plumes erupting from Europa's surface, further suggesting the presence of liquid water and the potential for exchange processes that could facilitate habitability. Additionally, the notion of tidal heating—caused by gravitational interactions with Jupiter and other moons—might play a role in generating conditions suitable for life by creating pockets of briny liquid within the ice shell.

A significant challenge for the Europa Clipper mission is navigating the intense radiation environment around Jupiter, particularly at Europa. The innovative design of the mission allows for reduced exposure to radiation by minimizing time spent in this hazardous area, ultimately lowering the spacecraft's complexity and costs compared to previous mission concepts.

Furthermore, the irradiation of Europa's icy surface is known to produce various oxidants, including molecular oxygen (O2), hydrogen peroxide (H2O2), carbon dioxide (CO2), and sulfur dioxide (SO2), although the mechanisms behind this production are not yet fully understood. As signs of potential life-sustaining conditions emerge, the mission will also work to characterize the radiation environment surrounding Europa and evaluate scientifically compelling sites for a potential future lander mission. This comprehensive approach aims to further our understanding of this fascinating moon and its capacity to host life, making it a pivotal point of focus in astrobiological research.

Microbial Habitability of Icy Worlds

(https://www.montana.edu/priscu/documents/Publications/PriscuandHand2012Habitability.pdf)

As our exploration of the cosmos nears its sixth decade, the inquiry into the habitability of icy worlds has gained considerable momentum, particularly focusing on the potential for microbial life. Throughout much of Earth’s history, life began as microscopic organisms, and even today, microorganisms comprise the predominant form of life on our planet. This understanding leads scientists to hypothesize that, if life were to exist beyond Earth, it likely originated as microbial life and may remain in that form. Consequently, contemporary astrobiological research increasingly emphasizes the metabolic capabilities of microorganisms and how they adapt to extreme environments.

A cornerstone of the search for extraterrestrial life is the quest for liquid water, as it has long been recognized that water is closely associated with life on Earth. However, the concept of where liquid water can be found is evolving, particularly concerning certain celestial bodies. Notably, several moons in the outer solar system—such as Europa, Ganymede, and Enceladus—are encased in ice but are believed to harbor vast subsurface oceans. These oceans could potentially hold several times more liquid water than exists on Earth, thus presenting some of the most extensive habitats available within our solar system for the pursuit of life.

The factors sustaining these subsurface oceans include radiogenic decay, which generates heat, and tidal energy dissipation caused by gravitational interactions with their parent planets. Importantly, the insulating properties of ice allow these oceans to remain in a liquid state, fostering conditions conducive to life. The existence of these liquid water bodies has likely persisted throughout much of the solar system’s history, providing key environments ripe for the search for life stemming from an independent origin.

Among these icy worlds, Europa stands out as a leading candidate for microbial habitability. Its ice shell is relatively thin, estimated to be between 5 to 15 kilometers thick, which may reduce the technical challenges involved in detecting life. Beneath this ice, Europa is believed to possess a rocky seafloor, which could provide both the energy and the essential chemical elements necessary to support metabolic processes.

In evaluating the habitability of such oceanic worlds, researchers focus on three core requirements: the presence of liquid water, a suite of biologically essential elements (such as carbon, nitrogen, and phosphorus), and a reliable source of energy. The subsurface ocean of Europa has likely remained stable for almost 4.5 billion years, making it a prime candidate for habitability. The geochemical interactions at the ocean floor, coupled with its vast volume, suggest that this environment could be suitable for supporting life, should the chemical conditions align favorably.

Alongside these fundamental criteria, other environmental parameters need to be considered, such as water activity, temperature, radiation levels, pH, salinity, and stability over time. A crucial aspect for habitability is the availability of free energy in the environment, which drives essential biological processes like metabolism, repair, and reproduction. Thus, the theme of "following the energy" complements the traditional astrobiological mantra of "following the water" in guiding searches for life in outer space.

The environment within Europa's ocean, while submerged under kilometers of ice, is not extraordinarily extreme in terms of temperature or pressure. Current models suggest that the temperature could be kept around the freezing point due to salt dissolution effects, while the pressures, although substantial, are less intense than those found in the deepest trenches on Earth, such as the Mariana Trench.

NASA's ongoing commitment to missions investigating Europa emphasizes the scientific importance of excavating possible microbial habitats beyond Earth. The prospect of discovering life on Europa or similar outer solar system bodies represents not only a significant scientific breakthrough but also a profound shift in our understanding of life's potential diversity across the universe. Such findings would reverberate through scientific domains and societal perspectives, reshaping our grasp of life's origins and its limits.

Searching for Life on Habitable Planets and Moons

(https://arxiv.org/pdf/0912.1040)

The paper titled "Searching for Life on Habitable Planets and Moons," authored by Ashwini Kumar Lal, delves into the intriguing quest for extraterrestrial life, underscoring the fact that Earth remains the only known inhabited planet in our vast universe. Recent advancements in astrobiology and observational astronomy, bolstered by exciting discoveries of extremophiles—organisms capable of thriving in the most inhospitable environments on Earth—have spurred speculation about the possibility of life existing on various celestial bodies.

The theory of panspermia, which posits that life may be distributed throughout the cosmos via meteors, asteroids, and comets, further supports the notion that life could potentially take root on planets and moons beyond our own. This theory suggests that "seeds of life" might be found intermixed within cosmic dust and debris, occasionally landing on Earth and other habitable locations in the universe.

Key targets for astrobiological exploration within our solar system include Mars, with its promising signs of past water; Europa, a moon of Jupiter believed to harbor a subsurface ocean; and Titan, Saturn's largest moon, which possesses a dense atmosphere and liquid methane lakes. Additionally, the icy moons Ganymede, Enceladus, and Callisto are all considered hotspots in the search for life.

Meanwhile, efforts to discover life beyond our solar system have gained momentum with missions like NASA's Kepler and the French CNES-ESA's CoRoT, both designed to identify Earth-like planets residing in the habitable zones of stars similar to our Sun. Kepler, in particular, has been meticulously observing approximately 150,000 stars, utilizing a method known as the transit technique to detect exoplanets. This approach allows astronomers to measure the size of an exoplanet by monitoring the dimming of a star's light when the planet passes in front of it.

The historic discovery of exoplanets began in 1995 with the identification of the first known extrasolar planet, '51 Pegasi b,' which orbits a G-type star in a remarkably close orbit. Over the years, the catalog of known exoplanets has grown significantly, highlighting the diverse and distinct types of these celestial bodies, including gas giants and super-Earths. Through these extensive research efforts, scientists are continually seeking answers to the profound question of whether we are alone in the universe.

On the Habitability of Europa

(https://doi.org/10.1016/0019-1035(83)90037-4)

The paper by Reynolds et al. investigates the possibility of habitability on Europa, one of Jupiter's four large Galilean moons. It posits that tidal and radiogenic heating have likely resulted in the formation and maintenance of a subsurface ocean of liquid water beneath a thin ice crust. The study examines the environmental factors that may support life, emphasizing that the proposed ocean could provide appropriate temperatures and long-term stability, which are crucial for the survival of known organisms.

The authors assert that the necessary biogenic elements for life are expected to be present, linked to the ocean's formation process. They identify various sources of biologically useful energy—such as electrical energy from hydrothermal vents, thermal energy from the moon's interior, and chemical energy from interactions between the ocean and the icy crust—as potentially critical for sustaining life on Europa.

Furthermore, they calculate the rates of resurfacing in an active crust model to estimate how much photosynthetically active radiation could penetrate the ice through fractures, allowing for potential biological activity beneath the surface. Drawing parallels from microorganisms found in Antarctic ecosystems, the authors discuss the possible biomass that could be supported by this energy.

While the authors acknowledge that their calculations cannot definitively establish the existence of life on Europa, they suggest that certain regions within the moon may exhibit conditions suitable for life, albeit in limited spatial and temporal contexts.

The paper also references compelling observations made during the Voyager missions, which revealed Europa's bright surface marked by a network of linear features that may indicate ongoing geological activity, such as fracturing of the ice crust. These features suggest that the moon is geologically active, possibly due to tidal flexing and internal processes, which could maintain the ocean and create environments conducive to life.

Overall, the research highlights the significance of Europa as a candidate for hosting life beyond Earth, emphasizing the need for further investigation of its icy surface and underlying ocean to understand its habitability better.

Astrobiology and the Potential for Life on Europa

(https://www.researchgate.net/profile/Kevin-Hand/publication/252433874_Astrobiology_and_the_Potential_for_Life_on_Europa/links/63e40f9f6425237563996dcf/Astrobiology-and-the-Potential-for-Life-on-Europa.pdf)

The text examines the possibility of life on Europa, one of Jupiter's fascinating moons. A key focus is Europa’s subsurface ocean, which is crucial for studying the potential for life beyond Earth. A team of researchers from NASA and various universities has identified important elements needed to support life, such as liquid water, essential biological components, and energy sources.

Europa stands out as a strong candidate for hosting life, both currently and in the past. Its rocky ocean floor and unique composition create conditions that may foster life. The subsurface ocean, believed to have existed for a long time, provides a stable environment where life could thrive. The interaction between the ocean and the ocean floor is thought to create a rich mix of chemicals that could support life.

When compared to other bodies like Mars and Enceladus, Europa's ocean is more stable. Mars may have had oceans in the past, but now it mostly contains temporary subsurface water. This lessens the chances of finding current life there. Concerns about contamination from Earth-like organisms also complicate the search for life on Mars. Enceladus shows geological activity that raises questions about the stability of its subsurface water, making it less reliable than Europa.

Exploring Europa could help answer important questions about life beyond Earth. Finding life there would change our understanding of what life is and how it evolves, suggesting that life could exist wherever conditions are right. Studying Europa’s ability to support life will provide valuable insights into the search for life in the universe.

The text also discusses how scientists might detect life on Europa by studying the relationship between its surface ice and the ocean underneath. The upper layers of ice are hostile to life, but there may still be signs of life if materials from the ocean rise to the surface. These signs, known as biosignatures, could prove that life existed there, either in the past or present.

To improve the chances of finding biological materials, researchers focus on areas with younger ice. These areas are more likely to have preserved traces of ancient life. Scientists classify biosignatures into two types: general biosignatures, which may indicate any form of life, and specific molecular markers, which are linked to biological processes such as DNA.

Additionally, the text points out that signs of past life, like fossilized remains, could be detected from orbit around Europa. Observations might reveal shapes or features influenced by biological activity, supporting the idea that life once existed. Recognizing unique molecular structures that indicate biological activity is also important. These molecules, and how they are arranged, could provide strong evidence that life may have existed on Europa. Through ongoing research and exploration, scientists hope to uncover the mysteries of this icy moon and its potential to support life.